EAS5450 Note on Venture Capital Method

(Classroom Example)

Introduction

Venture capital funds make investments in high growth companies with the expectation of high returns on those investments. While there is no one method that venture capitalists use to select, investigate, and value potential investment opportunities, the Venture Capital Method described below provides insight into the quantitative analysis of such investments.

> If you encounter unfamiliar financial terms when reading this note, there is an excellent glossary of venture capital terminology on the Investopedia, Pitchbook, and CooleyGO websites.

In this scenario, a venture capitalist is posing the following question:

If we invest a certain amount of capital in this company, and

- *we would like to see a specific return on the investment, and*
- *based on the company's business plan, we expect the company to grow in this way, and*
- *the company is valued in this way, and*
- *the company will have this liquidity event allowing me to realize my return on investment,*

Then:

- *What % or # of shares of the company should we own?*
- *What if there are subsequent rounds of investment that dilute our ownership?*
- *What happens to the entrepreneurs' and other investors' shares?*
- *What if things don't go as planned?*

VC Method: Single round

Investment

With a venture capital (VC) investment, an investment is made in a lump sum at a defined time (I_0) in exchange for shares of stock in the company. The return on investment is realized at a *liquidity event* sometime in the future.

> *Liquidity Event*: Specific event that allows the company's shares of stock to be exchanged for cash or another form of liquid security. Examples include:
>
> - Acquisition (for cash or stock of a publicly traded company)
> - Initial public offering (shares may be sold for cash in public markets)
> - Buyout (recapitalization that allows investors' shares to be bought back by the company or others for cash)

The venture capitalist may realize the return on investment by selling the shares of stock at a price determined by the value of the company at the time of liquidity event.

Company Value

There are many ways of valuing companies, many of which are as much art as science. One way to value a company is a multiple of revenues (R) or earnings (E) at a point of time (N). Assume a "success scenario," in which a company is sold (or has another form of liquidity event) in N years.

The "Terminal Value" of the company at year N may be calculated as:

$V_N = P/E \times E_N$ or

$V_N = P/R \times R_N$

where P/E and P/R are the price to earnings and price to revenue multiples, respectively.

> Price to earnings (P/E) and price to revenue (P/R), together with other multiples, may be derived by examining comparable companies in comparable industries as the company in question. Multiples must be calculated from comparable companies that have explicit market values, such as publicly traded companies.

EXAMPLE: Value of a company in year 5, based on income statement projections…

Sales projected in 5 years	$30MM
Cost of Goods Sold (COGS)	-18
Gross Margin	12
Sales, Marketing, R&D, G&A expenses	-7.5
Taxes	-2
Net Income (earnings)	**$2.5MM**

Referring to financial data for comparable companies, we find a P/E = 15.
$\therefore$ Value of the company in 5 years: $V_5 = 15 \times \$2.5MM = \$37.5MM$

Value of Investment

We have talked about two important concepts that are critical to Venture Capital:

- Time Value of Money: a dollar today is worth more than a dollar tomorrow
- Risk/Return trade-off: the higher the risk of an investment, the higher the expected return.

The figure below illustrates the Risk/Return tradeoff and provides one example of how later stage investment returns decrease as the risk is driven out of the business and value of the business increases.

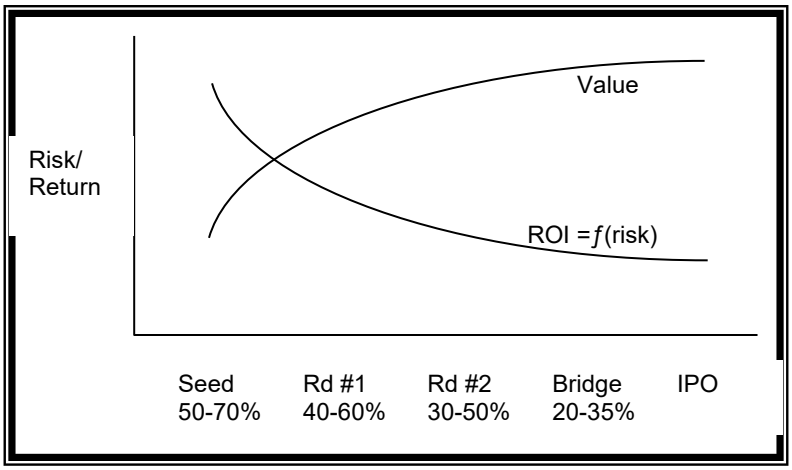


FIGURE: ***Risk/Return tradeoff in rounds of investment.***

A venture capital investor determines the target value of an investment after N years (VI_N) based on his/her desired annual Return on Investment (ROI):

$$VI_N = I_0 \times (1 + ROI)^N$$

EXAMPLE: Value of a $500,000 investment in year 5, given desired ROI of 50% per annum =

$$VI_5 = \$0.5MM \times (1 + 0.50)^5 = \$3.80MM$$

Alternatively, the investor may simply specify that he/she wants to receive "X" times their investment after N years, in which case:

$$VI_N = I_0 \times X$$

Ownership

We are now able to calculate the percentage of the company that the VC must own to realize the required ROI as well as the number of shares of stock.

Percent Ownership of Company to obtain required ROI

$$\%_N = \frac{(1+ROI)^N \times I_0}{P/E \times E_N}$$

EXAMPLE: Percent ownership required

In 5 years, at the liquidity event, the VC must own:

$$\%_5 = \frac{3.80}{37.5} \times 100\% = 10.13\% \text{ of the company.}$$

Now, we must determine the number of shares the VC should own. Rather than giving the VC shares of stock from the current owners/investors, companies will issue new shares of stock.

$$\%_{ownership} = \frac{\#\text{ of Shares Owned}}{\text{Total }\#\text{ Shares Issued}}$$

Total # Shares = Old # Shares (pre-investment) + New # Shares (purchased)

$$\%_{ownership} = \frac{\#New\ Shares}{\#Old\ Shares\ +\ \#New\ Shares}$$

$$\#\ \text{New Shares} = \frac{\%_{\text{ownership}}}{1 - \%_{\text{ownership}}} \times \#\ \text{Old Shares}$$

EXAMPLE: Number shares acquired and price per share

If 1,000,000 shares have been issued (e.g., founders' shares) prior to this first round of investment:

$$\therefore\ \#\ \text{New Shares} = \frac{0.101}{1 - 0.101} \times\ 1{,}000{,}000 = 112{,}350\ \text{shares}$$

$$\text{Price per share} = \frac{\text{Investment}}{\#\ \text{New Shares}} = \frac{\$0.5\text{MM}}{112{,}350\ \text{shares}} = \$4.45/\text{share}$$

Pre-Money and Post-Money Valuation

Post-Money Valuation, the value of the company after the VC investment has been made, is most commonly used to discuss the value of a company.

$$\text{Post - Money Valuation} = \frac{\text{Investment}}{\%_{\text{ownership}}}$$

or Post-Money Valuation = (New Price/Share) × (Total # Shares Post-Money)

Pre-Money Valuation = Post-Money Valuation – Investment

or Pre-Money Valuation = (New Price/Share) × (# Old Shares)

The Pre-Money Valuation represents the value of the company immediately prior to the investment.

EXAMPLE: Pre- and Post-Money Valuations

$$\text{Post - Money Valuation} = \frac{\$500{,}000}{0.101} = \$4{,}950{,}000$$

or \$4.45/share x (1,000,000 + 112,350) = \$4,950,000.

Pre-Money Valuation = \$4,950,000 – \$500,000 = \$4,450,000

or \$4.45/share x 1,000,000 = \$4,450,000.

In addition to shares of stock, several other securities may be involved in financing early stage companies.

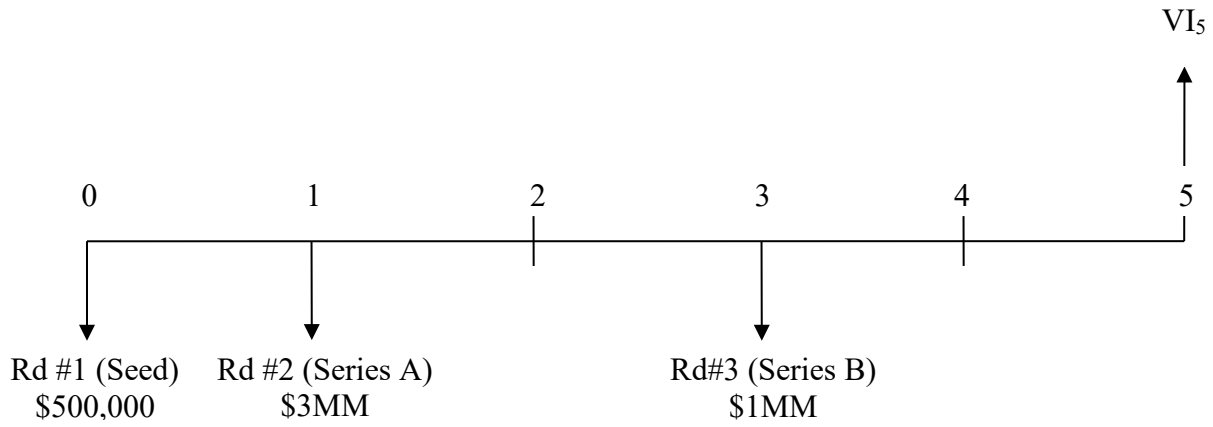
Options are the right to buy a certain number of shares of existing stock at a certain price in the future.

Warrants are similar to options except new stock is issued when warrants are exercised.

Valuations that include options and warrants as if they had already been exercised in the total number of shares are **fully diluted**.

VC Method: Multiple rounds

Typically, venture-backed companies have multiple rounds of investment. Consider the following example in which a company has three rounds of investment, as reflected in the figure below:



Percent Ownership

Let's assume that due to the gradual reduction of risk in the business, the required ROI for each round is progressively decreased:

Round	Invested	Target ROI	VI_N	Terminal Ownership*
#1 (Seed)	$0.5MM	50%	$3.80MM	10.13%
#2 (Series A)	$3.0MM	40%	$11.52MM	30.73%
#3 (Series B)	$1.0MM	25%	$1.56MM	4.17%

* Calculated based on formulas above.

NOTE: This would leave 55% for the founders, options, or other shares for future negotiations.

Retention Percentage

However, for the Round #1 investor to end up with their required 10.1% of the company, they must acquire more at the time of their investment. This is a result of the dilution that occurs in Rounds #2 and #3, decreasing the percentage of the company owned by previous shareholders as new shares are issued for each round.

$$\text{Retention \%} = 100\% - \sum(\text{Final \% of Later Rounds})$$

$$\%_{\text{Acquired}} = \frac{\text{Terminal \%}}{\text{Retention \%} \div 100\%}$$

Example: Retention percentage for 3 rounds

Round	Retention %
#1 (Seed)	100% - 30.73% - 4.17% = 65.10%
#2 (Series A)	100% - 4.2% = 95.83%
#3 (Series B)	100% = 100%

∴ At the time of their respective investments, each investor must acquire:

Round	% Acquired
#1 (Seed)	10.13% ÷ 0.651 = 15.55%
#2 (Series A)	30.73% ÷ 0.958 = 32.07%
#3 (Series B)	4.17% ÷ 1.0 = 4.17%

Number and Price of Shares

Now, that we have determined the retention percentage for each round's investor(s), we can complete the picture by calculating the number of new shares purchased with each round and the price per share using the following calculations (from above):

$$\text{\# New Shares Purchased} = \frac{\%_{\text{Acquired}}}{1 - \%_{\text{Acquired}}} \times \text{\# Old Shares}$$

$$\text{Price per share} = \frac{\text{Investment}}{\text{\# New Shares}}$$

Example: New shares issued at each round of investment

Round	# New Shares Purchased
#1 (Seed)	$\frac{0.1555}{1 - 0.1555} \times 1{,}000{,}000 = 184{,}173$ shares
#2 (Series A)	$\frac{0.3207}{1 - 0.3207} \times (1{,}000{,}000 + 184{,}173) = 559{,}027$ shares
#3 (Series B)	$\frac{0.0417}{1 - 0.0417} \times (1{,}000{,}000 + 184{,}173 + 559{,}027) = 75{,}791$ shares

NOTE: The new shares issued at each round must be added to the previous "old" shares when calculating the "old" shares of a subsequent round.

Example: Price per share at each round of investment

Round	Price Per Share
#1 (Seed)	= $500,000 ÷ 184,173 = $2.715
#2 (Series A)	= $3,000,000 ÷ 559,027 = $5.367
#3 (Series B)	= $1,000,000 ÷ 75,791 = $13.194

Post-Money Valuation

We can now calculate the Post-Money Valuation of the company at each investment round using either of the following formulas:

$$\text{Post - Money Valuation} = \frac{\text{Investment}}{\%_{\text{Acquired}}}$$

or Post-Money Valuation = (New Price/Share) × (Total # Shares Post-Money)

Example: Post-Money Valuations

Round	Investment	% Acquired	Price/Shr	Total # Shares Post-Money	Post-Money Valuation
#1 (Seed)	$500,000	15.55%	$2.715	1,184,173	$3.215 MM
#2 (Series A)	$3,000,000	32.07%	$5.367	1,743,200	$9.355 MM
#3 (Series B)	$1,000,000	4.17%	$13.194	1,818,991	$24.000 MM

Pre-Money Valuation

We can now calculate the Pre-Money Valuation of the company at each investment round using either of the following formulas:

Pre-Money Valuation = Post-Money Valuation – Investment

or Pre-Money Valuation = (New Price/Share) × (# "Old" Shares Pre-Money)

EXAMPLE: Pre-Money Valuations

Round	Post-Money Valuation	Investment	Price/Shr	# Shares Pre-Money	Pre-Money Valuation
#1 (Seed)	$3.215 MM	$0.5MM	$2.715	1,000,000	$2.715 MM
#2 (Series A)	$9.355 MM	$3MM	$5.367	1,184,173	$6.355 MM
#3 (Series B)	$24.000 MM	$1MM	$13.194	1,743,200	$23.000 MM

Terminal Price per Share

Finally, to complete the scenario, we must calculate the terminal price per share at the time of the liquidity event. In this case, we are using the value of the company at time of liquidity event and the total number of shares outstanding.

$$\text{Terminal Price per share} = \frac{\text{Value of Company}}{\text{Total \# Shares}}$$

Example: Final price per share

$$\frac{\$37.5MM}{(1{,}000{,}000 + 184{,}173 + 559{,}027 + 75{,}791)shares} = \frac{\$37.5MM}{1{,}818{,}991\ shares} = \$20.616/\text{share}$$

Back-Check

In order to check our work, we should confirm that the return on investment for each round does in fact equal our original targeted returns.

$$\text{ROI} = \left(\frac{\text{Price per share}_{\text{sold}}}{\text{Price per share}_{\text{purchased}}}\right)^{1/N} - 1$$

Example: Returns on investment for each round

Round	ROI
#1 (Seed)	$(\$20.616 \div \$2.715)^{1/5}-1 =$ **50% p.a.**
#2 (Series A)	$(\$20.616 \div \$5.367)^{1/4}-1 =$ **40% p.a.**
#3 (Series B)	$(\$20.616 \div \$13.194)^{1/2}-1 =$ **25% p.a.**

The results above do, in fact, match the desired returns sought by our investors in each round.

Convertible Preferred Stock

The discussion above pertains to issues of common stock. Venture financings often use another form of security known as convertible preferred stock. Preferred shares have "preference" over common shares and any cash distributions owed to preferred shareholders are paid from available funds **before** distributions are made to common shareholders. Preferred shareholders may also receive an annual dividend payment equal to a prescribed percentage of their investment. This dividend is usually "cumulative" (i.e., if the company cannot afford to pay the dividend in one year, any unpaid amount rolls over to the next year and is added to the dividend due in the next year). The dividend may either be "compounding" or "non-compounding" (i.e., treated as compound interest or simple, non-compounding interest). At the time of a liquidity event, each share of preferred stock may be converted (Convertible Preferred Stock or Shares) into a new share of common stock **and** the accumulated dividends are converted into additional shares of common stock often at the original price-per-share paid by the investor.

Example: Distribution on Convertible Preferred Stock

In the above example, suppose the investor in Round #2 (Series A) negotiated for Convertible Preferred Shares having a non-cash, cumulative, compounding dividend of 5% per annum. As shown above, this investor invests $3,000,000 four years prior to the planned liquidity event and receives 556,910 shares at a price of $5.39 per share. At the time of the liquidity event, the cumulative, compounding dividend would be worth:

$$\text{Dividend} = I\,[(1+i)^N - 1\,] = \$3{,}000{,}000\,[\,1.05^4 - 1\,] = \$646{,}520$$

At the original share price of $5.367, this would convert to:

$$\text{Additional Shares} = \frac{\$646{,}520}{\$5.367/\text{share}} = 120{,}462 \text{ shares}$$

In this case the Round #2 investor's total holdings at the liquidity event would be:

$$\text{Total shares owned} = 559{,}027 + 120{,}462 = 679{,}489 \text{ shares}$$

The Round #2 investor's distribution from the liquidity event would then be:

$$\text{Distribution} = (\text{Value of Company}) \times \frac{(\text{Total shares owned})}{(\text{Total \# shares issued})}$$

$$= \$37{,}500{,}000 \times \frac{679{,}489}{1{,}818{,}991+120{,}462} = \$13{,}138{,}000$$

<u>Note:</u> If the dividend had been cumulative, <u>non</u>-compounding, at the time of the liquidity event the dividend would have been worth:

$$\text{Dividend} = I \times i \times N = \$3{,}000{,}000 \times 0.05 \times 4 = \$600{,}000$$

$$\begin{aligned}\text{Return on investment (ROI)} &= [\,\text{Distribution / Investment}\,]^{1/N} - 1 \\ &= [\,\$13{,}138{,}000 / \$3{,}000{,}000\,]^{1/4} - 1 \\ &= 44.7\%\ \text{p.a. which, as expected, exceeds their 40\% target return.}\end{aligned}$$

Participating Preferred Stock

This security is like the Convertible Preferred Stock described above, but it has an additional benefit in that, at conversion (i.e., at the liquidity event), the entire original purchase price is **<u>also</u>** repaid to the investor on a priority basis before any other distributions to shareholders are calculated.

Example: Distribution on Participating Preferred Stock

In the above example, if only the Round #2 (Series A) investor has Participating Preferred Stock, then the investor's distribution at the time of the liquidity event would be:

$$\text{Distribution} = \text{I} + \left([(\text{Value of Company}) - \text{I}] \times \left[\frac{(\text{Total shares owned})}{(\text{Total \# shares issued})} \right] \right)$$

$$= \$3{,}000{,}000 + \left([\$37{,}500{,}000 - 3{,}000{,}000] \times \left[\frac{679{,}489}{1{,}939{,}453} \right] \right)$$

$$= \$15{,}087{,}000$$

Return on investment (ROI) = [Distribution / Investment]$^{1/N}$ – 1
= [\$15,087,000 / \$3,000,000]$^{1/4}$ – 1
= 49.75% p.a. which, as expected, exceeds their 40% target return.

Convertible Note

The convertible note is described in the EAS545 assigned reading, "Convertible Notes in Seed Financings," published by Harvard Business School Publishing, Item 9-813-017, Sept 23, 2016. Refer to this document if you are unfamiliar with this type of security.

A convertible note initially represents an interest-bearing loan from an investor to a start-up company. The loan converts to equity shares of the company when, and if, the company receives a subsequent (Series A) venture capital investment. Typical terms of a convertible note include:

- The principal amount of the note
- The note's interest rate (typ. 6-8% per annum compounded)
- Duration of the note (typ. 12-24 months)
- Minimum size of the next investment round that would trigger conversion
- Discount on Series A price/share for note conversion (typ. 20%)
- Pre-money valuation cap on conversion
- If a liquidity event (e.g., acquisition) occurs prior to a Series A financing, the note issuer receives a premium over the note principal
- Right to participate in the next financing round
- Seat(s) on the company's board of directors
- Consequences if the note duration is exceeded prior to a Series A financing

Conversion of the note is triggered by a Series A financing the amount, of which, is not less than the minimum threshold specified in the terms of the convertible note. Assuming the trigger threshold is met, to determine the pricing and number of shares acquired by the note's investor at the time of conversion (i.e., the time of Series A), we first determine the post-money value of the note as follows:

$$(\text{Post-Money Valuation})_{\text{Note}} = (\text{Pre-Money Valuation})_{\text{Series A}} \times (1 - \text{Discount})$$

Then, the percentage of the company acquired by the note's investor at the time of conversion is:

$$(\%_{\text{Acquired}})_{\text{Note}} = \frac{(\text{Principal+Accrued Interest on Note})}{(\text{Post-Money Valuation of Note})}$$

where:

$$(\text{Principal} + \text{Accrued Interest on Note}) = PI_N = (\text{Principal of Note}) \times (1 + \text{Note Interest Rate})^N$$

and N is the time between issuance of the note and the Series A investment.

The number of shares (S_{Note}) acquired by the note's investor at the time of conversion is then:

$$S_{\text{Note}} = \frac{\%_{Acquired}}{1 - \%_{Acquired}} \times (\text{\# Old Shares})$$

where # Old Shares is the number of shares issued prior to the convertible note (e.g., founders' shares).

And finally, the price per share (P_{Note}) acquired by the convertible note investor is:

$$P_{\text{Note}} = \frac{PI_{Note}}{S_{Note}}$$

We then must check to make sure the pre-money valuation cap of the convertible note has not been exceeded. As described in the article, "Convertible Notes in Seed Financings," the pre-money valuation cap is, by convention, assumed to pertain to the pre-money valuation of the convertible note, not to that of the Series A investment. Therefore, the maximum capped price of the convertible note shares is:

$$P_{\text{Note Cap}} = \frac{\text{Pre-Money Valuation Cap}}{\text{\# Old Shares}}$$

As long as $P_{\text{Note Cap}}$ is greater than P_{Note}, then S_{Note} and P_{Note} are valid. Otherwise, S_{Note} must be recomputed subject to the cap price.

Example: Convertible Note Seed Funding

Assume the Round #1 seed financing in the prior example is a convertible note subject to the following terms:

- Principal: $500,000
- Interest: 6% p.a. compounded
- Term: 18 months
- Minimum round to trigger conversion: $1,000,000
- Price discount on conversion: 20%
- Pre-money valuation cap: $6,000,000

As before, the Series A round occurs one year after the seed round, which is within the note's 18-month expiration term. The Series A investment of $3,000,000 exceeds the $1,000,000 minimum therefore this will trigger conversion of the note. The principal and interest on the convertible note at the time of the Series A round is then:

$$PI_N = \$500{,}000 \text{ x } (1.06)^1 = \$530{,}000$$

The pre-money valuation of the Series A round is $6,355,000 as found before. The post-money valuation of the note is then:

$$(\text{Post-Money Valuation})_{\text{Note}} = \$6{,}355{,}000 \text{ x } (1 - 0.20) = \$5{,}084{,}000$$

The percent of the company acquired by the convertible note investor at the time of conversion is then:

$$(\%_{\text{Acquired}})_{\text{Note}} = \frac{(\$530{,}000)}{(\$5{,}084{,}000)} = 10.43\%$$

The number of shares (S_{Note}) acquired by the note's investor at the time of conversion is then:

$$S_{\text{Note}} = \frac{0.1043}{1 - 0.1043} \text{ x } (1{,}000{,}000) = 116{,}385 \text{ shares}$$

And finally, the price per share (P_{Note}) acquired by the convertible note investor is:

$$P_{\text{Note}} = \frac{\$530{,}000}{116{,}385 \text{ shrs}} = \$4.55/\text{share}$$

This is within the pre-money valuation cap stipulated in the terms of the note:

$$P_{\text{Note Cap}} = \frac{\$6{,}000{,}000}{1{,}000{,}000} = \$6/\text{share}$$

so the $4.55/share price of the convertible note shares is valid.

In this situation where a convertible note precedes a Series A financing, these convertible note shares must be included along with the founders' shares when determining the number of "old" shares to use in the calculation of the number of Series A shares (pursuant to Page 6 of this document).

Round	# New Shares Purchased	Price/Share
#1 (Seed)	116,385 shares by conversion	$4.55
#2 (Series A)	$\frac{0.3207}{1-0.3207} \times (1{,}000{,}000 + 116{,}385) = 527{,}025$ shares	$\frac{\$3{,}000{,}000}{527{,}025} = \$5.69$
#3 (Series B)	$\frac{0.0417}{1-0.0417} \times (1{,}000{,}000 + 116{,}385 + 527{,}025) =$ 71,453 shares	$\frac{\$1{,}000{,}000}{71{,}453} = \$14.00$

Note: The $5.69 Series A price found here is consistent with the convertible note price of $4.55 which is discounted 20% from the Series A price.

The post-money and pre-money valuations of Series A and Series B remain unchanged from those found earlier.

To complete this example, we calculate the terminal price per share at the time of the liquidity event:

$$\frac{\$37.5MM}{(1{,}000{,}000 + 116{,}385 + 527{,}025 + 71{,}453)shares} = \frac{\$37.5MM}{1{,}714{,}863\ shares} = \frac{\$21.87}{share}$$

As before, in order to check our work, we should confirm that the return on investment for each round does in fact equal our original targeted returns.

Round	ROI
#2 (Series A)	$(\$21.87 \div \$5.69)^{1/4} - 1 =$ **40% p.a.**
#3 (Series B)	$(\$21.87 \div \$14.00)^{1/2} - 1 =$ **25% p.a.**

The results above do, in fact, match the desired returns sought by our Series A and Series B investors. We can also find the overall return on the convertible note investment:

$$\left(\frac{Terminal\ Distribution\ to\ Seed\ Investor}{Seed\ Investment}\right)^{1/5} - 1 = \left(\frac{\$21.87 \times 116{,}385\ shrs}{\$500{,}000}\right)^{1/5} - 1$$

$$= 38.5\%\ p.a.$$

Despite the discount on the conversion price of the note, this ROI on the note is less than that of the Series A investment. This is because the ROI on the note during the one-year period between this seed investment and the Series A investment is only 6% p.a., pursuant to the stipulated interest rate on the note prior to conversion. This low, one-year interest accrual reduces the overall 5-year ROI on the seed investment.

Post-Money SAFE

The Post-Money SAFE (Simple Agreement for Future Equity) is similar to the Convertible Note but it is not interest-bearing and has no preset expiration term. It initially represents an interest-free investment from an investor to a start-up company. The investment converts to equity shares of the company when, and if, the company receives a subsequent (Series A) venture capital investment. Typical terms of a Post-Money SAFE include:

- The principal amount of the SAFE investment
- Minimum size of the next investment round that would trigger conversion
- Discount on Series A price/share for SAFE conversion (typ. 20%)
- Post-Money Valuation Cap on conversion
- If a liquidity event (e.g., acquisition) occurs prior to a Series A financing, the SAFE issuer receives a premium over the SAFE principal
- Right to participate in the next financing round
- Seat(s) on the company's board of directors

Conversion of the SAFE is triggered by a Series A financing the amount, of which, is not less than the minimum threshold specified in the terms of the SAFE. Assuming the trigger threshold is met, to determine the pricing and number of shares acquired by the SAFE's investor at the time of conversion (i.e., the time of Series A), we first determine the post-money value of the SAFE as follows:

$$(\text{Post-Money Valuation})_{\text{SAFE}} = (\text{Pre-Money Valuation})_{\text{Series A}} \times (1 - \text{Discount})$$

<u>NOTE:</u> The Post-Money Valuation of the SAFE to be used in the following calculations is the lesser of the Post-Money Valuation calculated above or the Post-Money Valuation Cap provided in the terms of the Post-Money SAFE.

Then, the percentage of the company acquired by the SAFE's investor at the time of conversion is:

$$(\%_{\text{Acquired}})_{\text{SAFE}} = \frac{(\text{Principal Amount of the SAFE})}{(\text{Post-Money Valuation of SAFE})}$$

The number of shares (S_{SAFE}) acquired by the SAFE's investor at the time of conversion is then:

$$S_{\text{SAFE}} = \frac{\%_{Acquired}}{1 - \%_{Acquired}} \times (\text{\# Old Shares})$$

where # Old Shares is the number of shares issued prior to the SAFE (e.g., founders' shares, other SAFEs, issued and unissued option shares).

And finally, the price per share (P_{SAFE}) acquired by the SAFE's investor is:

$$P_{\text{SAFE}} = \frac{Principal\ Amount\ of\ SAFE}{S_{SAFE}}$$

Example: Post-Money SAFE Seed Funding

Assume the Round #1 seed financing in the prior example is a Post-Money SAFE subject to the following terms:

- Principal: $500,000
- Minimum round to trigger conversion: $1,000,000
- Price discount on conversion: 20%
- Post-Money Valuation Cap: $6,500,000

As before, the Series A round occurs one year after the seed round, which is within the SAFE's 18-month expiration term. The Series A investment of $3,000,000 exceeds the $1,000,000 minimum therefore this will trigger conversion of the SAFE. The pre-money valuation of the Series A round is $6,355,000 as found before. The post-money valuation of the SAFE is then:

$$(\text{Post-Money Valuation})_{SAFE} = \$6{,}355{,}000 \text{ x } (1 - 0.20) = \$5{,}084{,}000$$

This value is less than the Post-Money Valuation Cap of the SAFE and, therefore, will be used in the calculations that follow.

The percent of the company acquired by the SAFE's investor at the time of conversion is then:

$$(\%_{Acquired})_{SAFE} = \frac{(\$500{,}000)}{(\$5{,}084{,}000)} = 9.84\%$$

The number of shares (S_{SAFE}) acquired by the note's investor at the time of conversion is then:

$$S_{SAFE} = \frac{0.0984}{1 - 0.0984} \text{ x } (1{,}000{,}000) = 109{,}078 \text{ shares}$$

And finally, the price per share (P_{SAFE}) acquired by the SAFE's investor is:

$$P_{SAFE} = \frac{\$500{,}000}{109{,}078 \text{ shrs}} = \$4.58/\text{share}$$

In this situation where a SAFE precedes a Series A financing, these SAFE shares must be included along with the founders' shares when determining the number of "old" shares to use in the calculation of the number of Series A shares (pursuant to Page 6 of this document).

Round	# New Shares Purchased	Price/Share
#1 (Seed)	109,078 shares by conversion	$4.58
#2 (Series A)	$\frac{0.3207}{1 - 0.3207} \times (1{,}000{,}000 + 109{,}078) = 523{,}576$ shares	$\frac{\$3{,}000{,}000}{523{,}576} = \$5.73$
#3 (Series B)	$\frac{0.0417}{1 - 0.0417} \times (1{,}000{,}000 + 109{,}078 + 523{,}576) =$ 70,985 shares	$\frac{\$1{,}000{,}000}{70{,}985} = \$14.09$

Note: The $5.73 Series A price found here is consistent with the SAFE shares price of $4.58 which is discounted 20% from the Series A price.

The post-money and pre-money valuations of Series A and Series B remain unchanged from those found earlier.

To complete this example, we calculate the terminal price per share at the time of the liquidity event:

$$\frac{\$37.5MM}{(1{,}000{,}000\ +\ 109{,}078\ +\ 523{,}576\ +\ 70{,}985)shares} = \frac{\$37.5MM}{1{,}703{,}639\ shares} = \frac{\$22.01}{share}$$

As before, to check our work, we should confirm that the return on investment for each round does in fact equal our original targeted returns.

Round	ROI
#2 (Series A)	$(\$22.01 \div \$5.73)^{1/4}-1 =$ **40% p.a.**
#3 (Series B)	$(\$22.01 \div \$14.09)^{1/2}-1 =$ **25% p.a.**

The results above do, in fact, match the desired returns sought by our Series A and Series B investors. We can also find the overall return on the SAFE investment:

$$\left(\frac{Terminal\ Distribution\ to\ Seed\ Investor}{Seed\ Investment}\right)^{1/5} - 1 = \left(\frac{\$22.01 \times 109{,}078\ shrs}{\$500{,}000}\right)^{1/5} - 1$$

$$= 36.9\%\ p.a.$$

Despite the discount on the conversion price of the SAFE, this ROI on the SAFE is less than that of the Series A investment. This is because the ROI on the note during the one-year period between this seed investment and the Series A investment is 0% p.a., pursuant to the interest-free nature of the SAFE prior to conversion. This interest-free period of the SAFE reduces the overall 5-year ROI on the seed investment.